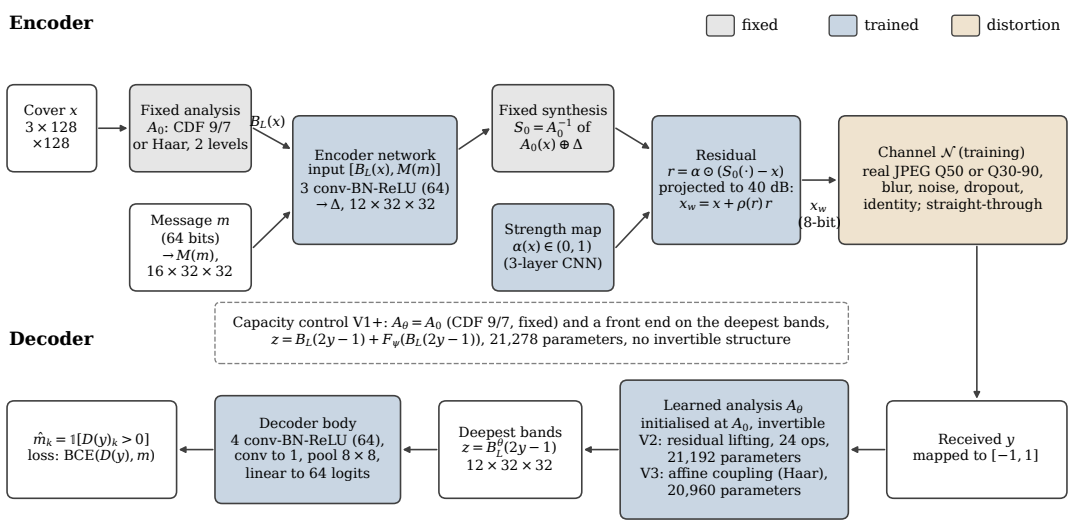
Encoder
fixed
trained
distortion
Cover x
3×128
$\times 128$
Fixed analysis
A_0: CDF 9/7
or Haar, 2 levels
$B_L(x)$
Message m
(64 bits)
$\to M(m)$,
$16 \times 32 \times 32$
Encoder network
input $[B_L(x), M(m)]$
3 conv-BN-ReLU (64)
$\to \Delta$, $12 \times 32 \times 32$
Fixed synthesis
$S_0 = A_0^{-1}$ of
$A_0(x) \oplus \Delta$
Strength map
$\alpha(x) \in (0, 1)$
(3-layer CNN)
Residual
$r = \alpha \odot (S_0(\cdot) - x)$
projected to 40 dB:
$x_w = x + \rho(r)\, r$
x_w
(8-bit)
Channel $\mathcal{N}$ (training)
real JPEG Q50 or Q30-90,
blur, noise, dropout,
identity; straight-through
Capacity control V1+: $A_\theta = A_0$ (CDF 9/7, fixed) and a front end on the deepest bands,
$z = B_L(2y-1) + F_\psi(B_L(2y-1))$, 21,278 parameters, no invertible structure
Decoder
Received y
mapped to $[-1, 1]$
Learned analysis A_θ
initialised at A_0, invertible
V2: residual lifting, 24 ops,
21,192 parameters
V3: affine coupling (Haar),
20,960 parameters
Deepest bands
$z = B_L^\theta(2y-1)$
$12 \times 32 \times 32$
Decoder body
4 conv-BN-ReLU (64),
conv to 1, pool 8×8,
linear to 64 logits
$\hat{m}_k = \mathbb{1}[D(y)_k > 0]$
loss: BCE$(D(y), m)$